

20250414EB_MN (NCT, ATP, TDP43, RNA)

Project: Yi-Shuan Tseng Holt Lab

Author: Yi Shuan Tseng

Status: In progress

Entry Created On: 06 May 2025 21:12:21 UTC

Entry Last Modified: 13 Jun 2025 05:18:40 UTC

Export Generated On: 01 Sep 2026 18:35:57 UTC

2025年4月14日 星期一

Use 2.5mM T8 full supp (made by no glucose ADMEM/F12 and Neurobasal-A) for differentiation and experiment. i didn't add extra Na pyruvate.

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2025年4月16日 星期三

T2 -> T3

2025年4月18日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年4月21日 星期一

T4 -> T5

2025年4月23日 星期三

T5 -> T6

2025年4月25日 星期五

T6 -> T6 (~14mL for over-weekend)

Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100)

2025年4月26日 星期六

Coating

1. Remove the PLO and wash the plate with PBS once
2. Coat the plate with laminine

2025年4月28日 星期一

T6 -> T7

Dissociation

1. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
2. Add trypsin (usually 3-4 mL/10-cm plate) and pipette it up and down with 5mL serological pipette.

3. Put the tube into 37C water bath. Pipette the cells with P1000 every 5 min round 8-10 times until 15 min. (3 times in total. There might be more resistance at the 2nd round)
4. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently. (Try to make the viscous cluster get through the tip to break it down)
5. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
6. Filter the cell with 40um filter into a new 50mL tube
7. Spin the cell down at 300rcf, 7min. Remove the supernatent.
8. Add T7 (~5mL) to resuspend the cells and count it.
9. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info ^

	A	B	C
1	cell	passage	well#
2	WT11	P16+1	5

Table1 ^

	A	B
1	run1	volume (mL)
2	trypsin	4
3	HI-FBS/CTWM	8

Table3 ✓

	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	1.61	100	5	8.05	A	6	1200	4	2981
3					C,D	24	232	14	the rest

plate A ^

	1	2	3
A			
B			

plate B (coated on 4/21) ^

	1	2	3	4	5	6
A						
B						
C						
D						

plate C ^

	1	2	3	4	5	6
A				PLO only +PBS		
B						
C	PLO_laminin					
D						

2025年4月29日 星期二

Change T7 -> T7 if added virus

2025年4月30日 星期三

T7-> T7 if not added virus

2025年5月1日 星期四

T7 -> T8

2025年5月2日 星期五

1/2 T8 -> 1/2 T8

2025年5月5日 星期一

1/2 T8 -> 1/2 T8

2025年5月7日 星期三

1/2 T8 -> 1/2 T8

Treat Plate A at 11:45 am for 24h groups.

plate A1				^
	1	2	3	
A	2.5m glu M 1h	0.15 mM glu 1h		
B	2.5mM glu 24h	0.15mM glu 24h		

2025年5月8日 星期四

PLATE A

Treat plate A for 1h groups at 11:08 am

RNA extraction at 12:09 am

wash cells once with cold PBS

add 1mL TriZol and incubate it for 5 min

collect Trizol and store them in -80

2025年5月9日 星期五

PLATE B

vLH111, vLH110

interval: 30s

Pre-activation: 3 loops

activation: 30 loops

Table2							^
	A	B	C	D	E	F	
1	condition #	glucose conc. (mM)	volume	FCCP (1000x)	2CG (333x)	SirDNA (5000x)	
2	1	2.5	2500			0.5	
3	2	0.15	2500			0.5	
4	3	0	2500			0.5	
5	4	0	2500	2.5	7.5	0.5	

Table5

	A	B	C	D	E	F
1			110 (B)	110 (B)	111 (C)	111 (C)
2	number	condition	treatment time	image 1h	treatment time	image 1h
3	1	2.5mM	12:05	13:00	9:05	10:05
4	2	0.15mM	12:40	13:46	9:30	10:36
5	3	0mM	13:20	14:31	10:00	11:12
6	4	0mM+FCCP+2DG	13:45	15:00	10:36	11:46

O.C. , 60X, 600x900, 12-bit

	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	640	12	30	300 ms	110_003 is 30% 100ms
3	561	12	10	100 ms	111_001 is 15%, 100ms
4	488	12	50	1s	

TDP43

Prepare 1st dilution: Halo ligand (stock: 1mM) to 2.5uM: $1000/2.5=400$.

Prepare 399uL DMSO with 1uL 1mM Halo dye.

positions: 10 soma, 5 neurite + movie (500ms, 40 loops)

Table4

	A	B	C	D	E
1	condition #	glucose conc. (mM)	volume	Halo ligand (2.5uM -> 1nM)	SPY595 (2500x)
2	1	2.5	1200	0.48	0.48
3	2	0.15	1200	0.48	0.48
4	3	0	1200	0.48	0.48

TDP43 exp time stamp					^
	A	B	C	D	
1	number	condition	treatment time	image 1h	
2	1	2.5mM	16:15	17:35	
3	2	0.15mM	16:45	18:05	
4	3	0mM	17:35		

O.C. , 60X, 704x704, 16-bit						^
	A	B	C	D	E	
1	channel	bit	power (%)	exposure time	note	
2	561	16	30	500ms		
3	640	16	40	300ms		
4	640 movie	16	40	500ms		

- note: I only see very few particles are moving... I need to double check the movie taken before.

vLH92
separate MN and non-MN
MN: 10 positions
non-MN: 8 positions

O.C. , 40X, zoom1.5, 600x900, 16-bit					^
	A	B	C	D	
1	channel	bit	power (%)	exposure time	
2	488	16	30	300	
3	Dapi-ex-Green-em	16	25	300	

treat three condition one after another. Run all condition at the same time.